BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – IST

Paper I (Objective Type) — Inter (Ist - A - Exam 2023)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6481

Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. Phenomenon of Isotopy was first discovered by :

(A) Millikan
(B) J. Perin
(C) Soddy
(D) J.J Thomson

Q2. The largest number of Molecules are present in ;

(A) 4.8 g of C_2H_5OH
(B) 3.6 g of H_2O
(C) 2.8 g of CO
(D) 5.4 g of N_2O_5

Q3. A Filtration Process could be very time consuming if it were not aided by a gentle suction which is developed :

(A) If the paper covers upto its circumference of funnel
(B) If the Paper has got small sized pores in it
(C) If Stem of the funnel is so large that it dips into the filtrate
(D) If the paper fits tightly

Q4. Temperature and quantity of a gas remains constant in :

(A) Charles's Law
(B) Avogadro's Law
(C) Boyle's Law
(D) Dalton's Law

Q5. S.I. Unit of Pressure is :

(A) Torr
(B) mm of Hg
(C) Pound inch^-2
(D) Nm^-2

Q6. Acetone and Chloroform are soluble in each other due to :

(A) Instantaneous Dipole
(B) Ion Dipole Interaction
(C) Intermolecular Hydrogen Bonding
(D) London Dispersion Forces

Q7. In Order to raise the boiling point of water upto 110 deg C, the external pressure should be :

(A) Between 760 Torr and 200 Torr
(B) Between 760 Torr and 1200 Torr
(C) 765 Torr
(D) Any Value of Pressure

Q8. The nature of the positive rays depend on :

(A) Nature of Anode
(B) Nature of Cathode
(C) Nature of the Residual Gas
(D) Nature of Discharge Tube

Q9. Which of the given do not obey Octet Rule :

(A) CH4
(B) NH3
(C) BCl3
(D) H2O

Q10. For the reaction NaOH + HCl -> NaCl + H2O the change in Enthalpy called :

(A) Heat of Reaction
(B) Heat of Neutralization
(C) Heat of Combustion
(D) Heat of Formation

Q11. The term which is not a State Function :

(A) Volume
(B) Enthalpy
(C) Work
(D) Internal Energy

Q12. The reaction for synthesis of NH3, the value of delta n is : N2 + 3H2 <=> 2NH3

(A) +2
(B) -2
(C) +1
(D) +4

Q13. An excess of Aqueous Silver Nitrate is added to aqueous Barium Chloride and precipitate is removed by filtration. What are main irons in Filtrate :

(A) Ag+ and NO3-
(B) Ag+, Ba2+ and NO3-
(C) Ba2+ and NO3-
(D) Ba2+, NO3- and Cl-

Q14. A solution of 10 % w/v of Glucose, the volume in which its 1 gram mole is dissolved will be :

(A) 1 dm3
(B) 1.8 dm3
(C) 200 cm3
(D) 900 cm3

Q15. The Oxidation Number of Sulphur in SO4^2- is :

(A) 4 (B) 3

(C) 6 (D) 0

Q16. If Salt Bridge is not used between two Half Cells then Voltage :

(A) Decrease Rapidly (B) Decrease Slowly

(C) Drops to Zero (D) Does not change

Q17. With Increase of 10 deg Celsius temperature rate of reaction double, this increase of rate is due to :

(A) Decrease in Activation Energy (B) Decrease in the Number of Collision between Reactants Molecules

(C) Increase in the number of effective collisions (D) Increase in Energy of Activation

Chemistry (Subjective) — Inter (Ist - A - Exam 2023)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: It is compulsory to attempt any (8 - 8) Parts each from Q.No. 2, Q.No.3 and attempt any (6) Parts from Q.No.4. Attempt any (3) Questions from Part - II. Write same Question No. and its Part No. as given in the Question Paper — (Part - I) — 22 x 2 = 44 — Make Diagram where necessary.

Q2.

Q.No.2

(i) Magnesium Atom is twice heavier than Carbon Atom. Justify the statement.

(ii) What is Critical Temperature of a Gas? What is its importance for Liquefaction of Gases?

(iii) What is Molecular Ion? Give an example.

(iv) What are Isotopes? Give example

(v) What are Natural and Artificial Plasma?

(vi) Define Boyle's Law. Give its Mathematical Expression.

(vii) Why is it necessary to decrease the pressure in the discharge tube to get the Cathode Rays?

(viii) Why e/m value of the Cathode Rays is just equal to that of Electron?

(ix) What are defects of Bohr's Atomic Model?

(x) Define Lattice Energy and give example.

(xi) State First Law of Thermodynamics and give its Mathematical Form.

(xii) What is State and State Function? Differentiate.

Q3.

Q.No.3

(i) Differentiate between Molality and Molarity.

(ii) Why is the Aqueous Solution of Ammonium Chloride Acidic?

(iii) What is meant by Water of Crystallization? Give two examples.

(iv) What are Pseudo First Order Reactions? Give an example.

(v) What do you mean by Inhibitor? Give an example.

(vi) Define Half Life Period. How is it related to order of reaction?

(vii) How can the decolourization of undesirable colours be carried out for freshly prepared crystals?

(viii) What is Solvent Extraction? Give its importance.

(ix) What is Sintered Glass Crucible? Give its significance.

(x) Evaporation causes cooling. Give the reason.

(xi) What are Dipole induced Dipole Forces?

(xii) Define Polymorphism. Give an example.

Q4.

Q.No.4

(i) Size of an Anion is always greater than that of its Parent Atom. Justify.

(ii) How bond length is affected by change in Hybridization state?

(iii) Why He2 does not exist under Normal Condition?

(iv) Justify that Chemical Equilibrium is dynamic in nature.

(v) Why do we need buffers in daily life?

(vi) How some reactions are effected by change in Pressure?

(vii) Na and K can displace Hydrogen from Acids but Pt, Pd and Cu can not? Explain

(viii) Lead Accumulator is chargeable battery Justify.

(ix) How reactivity of Metals is studied with the help of Electrochemical Series?

Note: (Part - II) — 3 x 8 = 24

Q5. (Marks 8) (a) Define Stoichiometry. Give its assumptions. Mention two important laws which help to perform the Stoichiometric calculation. (1+2+1 = 4) (b) Calculate the Mass of 1 dm3 of NH3 Gas at 30 deg C and 1000 mm Hg pressure, considering that NH3 is behaving ideally. (4)

Q6. (Marks 8) (a) Define Hydrogen Bonding and explain its any three applications. (1+3 = 4) (b) State and explain first law of Thermodynamics. (1+3 = 4)

Q7. (Marks 8) (a) Describe any four properties of Cathode Rays. (4) (b) What is the Percentage Ionization of Acetic Acid in a Solution in which 0.1 Moles of it has been dissolved per dm3 of the solution? (% Ionization = 1.3) (4)

Q8. (Marks 8) (a) Discuss the shapes and geometry of CH4 and H2O with reference to sp3 Hybridization. (4) (b) Write only four industrial applications of Electrolytic Process. (4)

Q9. (Marks 8) (a) Give Graphical Explanation for Elevation of Boiling Point of a Solution. (4) (b) How Rate of Reaction depends upon the following factors: (i) Nature of Reactants (ii) Surface Area (4)

GROUP – 2ND

Paper I (Objective Type) — Inter (Ist - A - Exam 2023)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6482

Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. The largest number of Molecules are present in

(A) 3.6 g of H2O
(B) 4.8 g of C2H5OH
(C) 2.8 g of CO
(D) 5.4 g of N2O5

Q2. Isotopes differ in :

(A) Properties which depends upon mass
(B) Arrangement of Electrons in Orbitals
(C) Chemical Properties
(D) The extent to which they may be affected in electromagnetic field

Q3. The colour of iodine in CCl4 solution is :

(A) Brown
(B) Purple
(C) Grey
(D) Black

Q4. The deviation of Gas from Ideal Behaviour is maximum at :

(A) -10 deg C and 5.0 atm
(B) -10 deg C and 2.0 atm
(C) 100 deg C and 2.0 atm
(D) 0 deg C and 2.0 atm

Q5. Graph between Pressure and Volume at constant temperature is called :

(A) Isobar
(B) Isochor
(C) Isotherm
(D) Spectrograph

Q6. Diamond is bad conductor because :

(A) It has tight structure
(B) It has high density
(C) Is Transparent to Light
(D) There are no free electrons present in crystal of diamond to conduct electricity

Q7. Which of the following Solid is an example of Covalent Solid with layered structure :

(A) Diamond
(B) Silicon Carbide
(C) Aluminium Nitride
(D) Graphite

Q8. Quantum Number value for 2p Orbital is :

(A) n = 2, l = 1
(B) n = 1, l = 2
(C) n = 1, l = 0
(D) n = 2, l = 0

Q9. Which of following Hydrogen Halide has highest percentage of Ionic Character :

(A) HCl
(B) HBr
(C) HF
(D) HI

Q10. For a given process, heat changes at constant pressure (qp) and heat changes at constant volume (qv) are related to each other :

(A) qp = qv
(B) qp > qv
(C) qp < qv
(D) qp = qv/2

Q11. Which of the given is not a State function :

(A) Heat
(B) Volume
(C) Pressure
(D) Enthalpy

Q12. For an Exothermic Reversible reaction, Increase in temperature will favour which :

(A) Forward Direction
(B) Reverse Direction
(C) Equilibrium will not disturbed
(D) Initially in forward direction, then in reverse direction

Q13. An excess of AgNO3 is added to Aqueous Barium Chloride and precipitate is removed by filtration. What are the main ions in Filtrate :

(A) Ag+ and NO3- only
(B) Ag+ and Ba2+ and NO3-
(C) Ba2+ and NO3- only
(D) Ba2+ and NO3- and Cl-

Q14. A solution of glucose is 10 % w/v, the volume in which 1 g mole of it is dissolved will be :

(A) 1 dm3
(B) 1.8 dm3
(C) 200 cm3
(D) 900 cm3

Q15. If a Salt Bridge is not used between two Half Cells then Voltage :

(A) Decreases Rapidly
(B) Decreases Slowly
(C) Does not change
(D) Drops to Zero

Q16. Which of the following product is obtained at Cathode during electrolysis of aqueous solution of Sodium Chloride :

(A) Na
(B) Cl2
(C) H2
(D) O2

Q17. With the Increase of 10 deg C temperature, the rate of reaction doubles. This increase in rate of reaction is due to :

(A) Decrease in Activation Energy of reaction
(B) Decrease in the Number of Effective Collisions between Reactant Molecules
(C) Increase in activation energy of reactants
(D) Increase in number of effective collisions

Chemistry (Subjective) — Inter (Ist - A - Exam 2023)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: It is compulsory to attempt any (8 - 8) Parts each from Q.No. 2, Q.No.3 and attempt any (6) Parts from Q.No.4. Attempt any (3) Questions from Part - II. Write same Question No. and its Part No. as given in the Question Paper — (Part - I) — 22 x 2 = 44 — Make Diagram where necessary.

Q2.

Q.No.2

(i) Differentiate between Actual Yield and Theoretical Yield.

(ii) N2 and CO have same number of Electrons, Protons and Neutrons. Justify it.

(iii) Define term Atomicity. Give example.

(iv) Why rate of diffusion of NH3 gas is more than HCl gas?

(v) Derive Boyle's Law from Kinetic Molecular Theory.

(vi) Differentiate between Continuous Spectrum and Line Spectrum.

(vii) Write two uses of Plasma.

(viii) What is Zeeman's Effect?

(ix) Cathode Rays are charged Particles. Justify.

(x) What is the Physical Significance of equation delta H = qp?

(xi) Differentiate between System and Surrounding.

(xii) State Ist Law of Thermodynamics. Give its mathematical equation.

Q3.

Q.No.3

(i) Define Solubility and Solubility Curves.

(ii) Give two applications of Colligative Properties.

(iii) What is Hydrolysis? Give an example.

(iv) What do you mean by Catalyst for a Catalyst? Give one example.

(v) How surface area affect the rate of a Chemical Reaction?

(vi) Define Rate of Reaction and give its Mathematical Expression.

(vii) Evaporation takes place at all temperatures. Explain with reason.

(viii) Why Methane is a gas while Hexane is a Liquid?

(ix) Define Isomorphism with an example.

(x) Define Sublimation with two examples.

(xi) How Decolouration of undesirable colour is done for crystals in Crystallization?

(xii) Why concentrated HCl and KMnO4 solutions can be filtered by Gooch Crucible?

Q4.

Q.No.4

(i) The bond angles of H2O and NH3 are not 109.5 deg like that of CH4. Although O- and N-atoms are sp3 Hybridized. Justify.

(ii) The radius of an Atom cannot be determined precisely. Give reason.

(iii) Differentiate between Sigma and Pi Bond.

(iv) Why solubility of Glucose in water is increased by increasing the temperature?

(v) Write equilibrium constant expression of the following reaction: N2(g) + 3H2(g) <=> 2NH3(g)

(vi) Differentiate between Reversible and Irreversible Reactions.

(vii) A Salt Bridge maintains the electrical neutrality in the cell. Give the reason.

(viii) Calculate the Oxidation Number of Chromium in the given compounds: (i) K2CrO4 (ii) Cr2O7^-2

(ix) Write the function of Salt Bridge.

> Note: (Part - II) — 3 x 8 = 24

Q5. (Marks 8) (a) What is Empirical Formula? Discuss steps to calculate Empirical Formula. (4) (b) What pressure is exerted by mixture of 2.00 g of H2 and 8.00 g of N2 at 273 K in 10 dm3 Vessel? (4)

Q6. (Marks 8) (a) How does Hydrogen Bonding explain the following indicated properties of the substances: (i) Hydrogen Bonding in Proteins (ii) Formation of Ice and its lesser density than Liquid Water (4) (b) State Hess's Law of Constant Heat Summation. Give two examples. (4)

Q7. (Marks 8) (a) Describe an experiment for the measurement of e/m value of electron. Also draw the diagram. (4) (b) When 1.00 Mole of steam and 1.00 mole of Carbon Monoxide are allowed to reach equilibrium 33.3 % of the equilibrium mixture is Hydrogen. Calculate the value of Kp. State the units of Kp. (4)

Q8. (Marks 8) (a) What is meant by Atomic Orbital Hybridization? Explain its one type in detail. (4) (b) Discuss Fuel Cells. Also give chemical equations of these fuel cells. (4)

Q9. (Marks 8) (a) Define Solubility Curves. Discuss Solubility Curves of NaCl and Ce2(SO4)3 (4) (b) Differentiate between Homogeneous and Heterogeneous Catalysis. Give two examples of each. (4)